

The distributive law



1 The student should multiply the second part of the sum, 2, by the number outside the brackets, 6 to get 12 as the second term on the right hand side.

2

a $5s + 35$

c $3r - 18$

e $-8t - 80$

g $-y - 2$

i $-80 - 30r$

k $pq + 4p$

m $-2m + 3n$

o $8r + 12s + 20t$

b $45 + 9w$

d $7y + 56$

f $-8c + 40$

h $16s + 20$

j $16t + 24$

l $9u - 9v$

n $-11rs - 13rt$

p $24m + 18n + 30p$

3

a $12mn + 18m$

c $r^2 + 7r$

e $-12a^2 - 54a$

g $16pq - 12p + 24pr$

i $-10y^2 - 70$

k $rs^2 + 8r$

m $16s^5t^4 - 36s^5$

o $9f^2 - 36fg - 54fh$

b $9v^2 + 54$

d $24m^2 + 42m$

f $4wy^2 + 4w^2y$

h $2f^2 - 14fg - 12fh$

j $-9 + s^3$

l $-30 - 80u^3$

n $12p^3q^5 - 54p^3$

p $3uv^2 + 3u^2v$

4

a $4x + 30$

e $36x - 11$

i $10y - 49$

b $15x + 27$

f $-11x + 24$

j $20x - 52$

c $3x + 17$

g $-x - 15$

d $18x + 32$

h $-40x + 47$

5

a $x^2 + 12x + 72$

e $13x - 16$

i $14u^2 + 20uv$

b $15y^2 - 39y$

f $4y + 44$

j $62w^2 + 34wz$

c $10x - 33y$

g $-11x - 26$

d $12y + 93$

h $17a + 20$

Applications



7 $(16r - 24)$ representatives

8 $(10 + 15n^4)$ boxes

9 $(16s^2t + 16st^2)$ units²

10 $(u^2 + 8u)$ m²

Factorisation

11

a Yes

b No

c Yes

d Yes

e No

f Yes

12

a 2

b 3

c 3

d 2

e 7

f $4x$

g 5

h 6

i 3

13

a 2

b 5

c 3

d 2

e $v + 3$

f $a - 4$

g $8a + 3$

h $2w - 9$

i 4

j 5

k 6

l $r + 7$

m $t - 2$

n $9u + 2$

14

a $6(v + 5)$

b $6(a - 8)$

c $5(9t - 8)$

d $2(m + 5n)$

e $6(9 - 10fg)$

f $8(g - 9h)$

g $3(3f + 5g)$

h $3(w + 5)$

i $2(c - 9)$

j $5(8c - 7)$

k $6(w + 5y)$

l $2(7y + 2w)$

m $2(2a - 7b)$

n $4(8 - 9jk)$

o $2(4jkt - 9)$

p $f(3g - 10h)$

q $rw(3t - 8)$

r $2rt(2 - 3r)$

s $4wy(4 - 9w)$

15 $2w - 10z + 8y = 2(w - 5z + 4y)$

16



a $4(2g + 7h - 5)$

c $4(u - 3v + 5w)$

e $3(m - 2n + 4p)$

g $z(w - y - k)$

i $4m(3 - 8n - 5mn)$

k $5rt(5r - 3t - 9rt)$

b $4(w + z + y)$

d $5(2m - 5n - 3p)$

f $5(2g - 7h + 3)$

h $5r(12 - 5t - 7rt)$

j $4z(4 - 8y - 6y^2)$

Factorisation of negatives

17

a $-(4r + 3s)$

b $-7(r + s)$

c $-(2 + 9r)$

d $-(8x + 5y)$

18

a $-8(b + 2)$

b $-2(9a - 8)$

c $-10(8v + 3)$

d $-6(6 + x)$

e $-5(5u + 7v)$

f $-7(5m - 7n)$

g $-2pr(4q + 3)$

h $-w^2(8 - 3y)$

i $-3j^2(8 + 9k)$

j $-m(8m - 9n)$

k $-uv(10u - 9v)$

l $-3j^2(7 + 12k)$

m $-2(s + 5)$

n $-8(v^2 - 7)$

o $-2(6s - 5)$

p $-7(4b + 3)$

q $-7(9r^2 - 4)$

r $-4(m + 8n)$

s $-7(7y + 3w)$

19

a $-9(2p + q + 5r)$

b $-5(2r + 5s - 4t)$

c $-5(4u + 5v - 9w)$

d $-2(4w + 9y + 5z)$